

CH9329 HID (Mouse & Keyboard) Module

Hardware Overview

What is it?

The **CH9329 Module** is a serial-to-HID (Human Interface Device) converter. When you plug it into a computer via USB, the computer believes you have plugged in a standard mouse and keyboard.

Electrical Explanation

- Simulation: The module does not have physical keys or a laser sensor. Instead, it listens for serial commands from K-Comp and translates them into standard USB HID packets.
- The "Unbrick" Mode: The CH9329 chip has several modes. It *must* be in **Mode 0** (Standard Mouse & Keyboard) to work with this library.
- Baud-Rate: The CH9329 *must* be configured to **9600 baud**. If your module was altered to 115200 baud it will not understand the commands sent by this library.

Key Hardware Facts:

- **RX Pin**: The K-Comp receives data on for example Digital Pin **D6**.
- **TX Pin**: The K-Comp sends commands on for example Digital Pin **D7**.
- **Baud-Rate**: 9600
- **Voltage**: Operates at 5V.
- **Signal Flow**: K-Comp → Serial Data → CH9329 Chip → USB Cable → Target Computer.

Software & Programming

kcomp_CH9329_hid.h

Parsing the raw text from a GPS device is difficult. This library handles reading, parsing, and math.

Usage: `#include <kcomp_CH9329_hid.h>`

The Hidden Key Map

How do you tell a computer to press F12, the Windows Key, or Enter? These aren't normal letters. They are raw hexadecimal numbers. Open `kcomp_CH9329_hid.h`! You will find a list of `#define` macros mapping out every key (e.g., `#define KEY_RETURN 0xB0`).

Function Reference

Function Name	Description & Parameters
<code>void beginCH9329(int rxPin, int txPin)</code>	Setup: Starts the module. ▪ <code>rxPin</code> / <code>txPin</code>: The pins connected to the module.
<code>void configureMode0()</code>	Repair: Forces the chip into the correct operating mode (Mode 0). Only use this if the computer stops recognizing the chip as a keyboard. <i>Wait 3 seconds after calling this for the chip to reboot.</i>

<code>void typeText(const char* text)</code>	Action: Automatically types out a full string. ▪ Example: <code>typeText("Hello!");</code>
<code>size_t writeKey(uint8_t k)</code>	Action: Presses and releases a single key. ▪ Example: <code>writeKey('a');</code>
<code>size_t pressKey(uint8_t k)</code>	Action: Presses and holds a key down. Useful for modifiers (Shift, Ctrl, Alt, Win/GUI).
<code>size_t releaseKey(uint8_t k)</code>	Action: Releases a key that was held down.
<code>void releaseAll()</code>	Action: Lifts all fingers off the fake keyboard. Always call this after using <code>pressKey()</code> .
<code>void moveMouse(int8_t x, int8_t y)</code>	Action: Moves the mouse relative to its position. ▪ x, y: Values from -128 to +127. Negative values = Left/Up. Positive = Right/Down.
<code>void moveMouseAbsolute(uint16_t x, uint16_t y)</code>	Action: Moves the mouse to a specific coordinate. ▪ x, y: Values usually range from 0 to 4095.
<code>void clickMouse(uint8_t buttons)</code>	Action: Presses mouse buttons. ▪ 1 = Left, 2 = Right, 4 = Middle, 0 = Release.
<code>void scrollMouse(int8_t wheel)</code>	Action: Scrolls the mouse wheel. ▪ 1 = Up, -1 = Down.

⚠ Pitfalls & Troubleshooting

- **The "Stuck Key" Trap** If you use `pressKey(KEY_LEFT_CTRL)` but forget to call `releaseAll()`, the target computer will think the Control key is permanently pressed!
The Fix: Always, always call `releaseAll()` after executing a macro combination.
- **The Device is Not Recognized**
If the computer says "USB Device Not Recognized," the chip might be in a different communication mode.
The Fix: Call `configureMode0()` once in your `setup()`, wait 3 seconds, and unplug/replug the USB cable.

Sample Program

This program waits for you to type a number into the Serial Monitor, and then executes a macro or mouse movement on the target computer.

```
#include <kcomp_CH9329_hid.h>

#define RX_PIN 6
#define TX_PIN 7

void setup() {
  Serial.begin(115200);
  beginCH9329(RX_PIN, TX_PIN);

  Serial.println("--- CH9329 HID Controller ---");
  Serial.println("Type '1' to draw a square with the mouse.");
  Serial.println("Type '2' to type a message.");
}

void loop() {
  if (Serial.available() > 0) {
```

```
char cmd = Serial.read();
while(Serial.available()) Serial.read(); // Clear buffer

if (cmd == '1') {
    Serial.println("Moving Mouse...");
    for(int i=0; i<10; i++) { moveMouse(10, 0); delay(20); } // Right
    for(int i=0; i<10; i++) { moveMouse(0, 10); delay(20); } // Down
    for(int i=0; i<10; i++) { moveMouse(-10, 0); delay(20); } // Left
    for(int i=0; i<10; i++) { moveMouse(0, -10); delay(20); } // Up
}
else if (cmd == '2') {
    Serial.println("Make sure a text window is active!");
    delay(2000); // Give user 2 seconds to click into a text box
    typeText("Hello from the K-Comp!");
    writeKey(KEY_RETURN); // Press Enter
}
}
```

 **Try it out:** You can find the full sample program **SimpleCH9329_HID_MouseKeyboard** in the K-Comp examples folder!